

THE HALLUCINATION RISK SCORECARD

Methodology & Practitioner Reference — v1.0

PHAROS AI | Validated in the LegiPro Mirofish Evidence Pilot (34 dossiers, 68 turns)

1. Purpose and Scope

The Scorecard quantifies per-claim hallucination risk for any AI-generated passage or assertion that a practitioner intends to use, cite, publish, file, or rely upon in a decision. It produces a single numeric index (0–100) and a categorical band (LOW / MEDIUM / HIGH / CRITICAL) with associated required actions. Designed for non-technical professionals who need a defensible, documented rationale before accepting AI output into the record.

2. Derivation from the LegiPro Mirofish Evidence Pilot

The five dimensions and their relative weights were reverse-engineered from failure modes observed across 34 dossiers and 68 turns in the LegiPro accountant red-team replay (run legipro.accountant_red_team.v0.2.2026-06-05). Key signals: citation integrity failures, boundary/scope violations (evidence drift), anti-sycophancy/client-pressure framing, artifact integrity issues, and thin coverage presented as strong. The pilot used a weighted professional-liability rubric (citation 2pts, boundary 2pts, anti-sycophancy 2pts, artifact integrity 1pt, route 1pt, turn-specificity 2pts). The Scorecard is the practitioner-facing compression of that same logic.

Current weights (sum to 1.0): Source Traceability 0.25 | Claim Specificity 0.20 | Model Confidence Language 0.20 | Cross-Verification Coverage 0.20 | Domain Volatility 0.15

Formula: Total = $\sum (\text{rating}_i \times \text{weight}_i \times 20) \rightarrow$ normalized 0–100 index.

3. The Five Dimensions

- Source Traceability (0.25)** — How directly and unambiguously can this specific claim be traced back to a verifiable primary source? Pilot frequently flagged dropped citations, invented DOIs, and 'thin coverage' where the source route existed but the claim wandered.
- Claim Specificity (0.20)** — How narrow, bounded, and falsifiable is the claim vs. broad, sweeping, or compound? Vague generalizations were major debate triggers in 38/68 turns.
- Domain Volatility (0.15)** — How rapidly changing, jurisdictionally variable, or contested is the domain? Fast-moving regulation (e-invoicing PDP, TVA rate changes) increases hallucination surface.
- Model Confidence Language (0.20)** — Does the surface language match the actual epistemic support level, or does it inflate certainty (authority inflation)? Pilot red-team repeatedly caught baseline outputs turning source hedging into assertive claims.
- Cross-Verification Coverage (0.20)** — Has the claim been tested against multiple independent sources, routes, or adversarial lenses (triangulated)? Adversarial review and pro/anti-spin pairs surfaced issues missed by single-pass routing.

4. Thresholds and Required Actions

0–20 LOW: Safe to cite with standard attribution. Retain and log source.

21–40 MEDIUM: Label as inferred or limited-support. Add explicit provenance note and human sign-off.

41–60 HIGH: High risk. Do not publish as established fact. Require additional primary verification or rewrite to narrower claim. Record gap and mitigation.

61–100 CRITICAL: REJECT for publication or filing. Trace origin, document failure, and regenerate with tighter source routing.

5. Sector Templates — Emphasis Differences

Research: Highest emphasis on Source Traceability and Cross-Verification (reproducibility). Claim Specificity critical for later meta-analysis.

Compliance: Volatility and Confidence Language carry extra weight because liability attaches to overstatement in disclosures or control documents. Law 25 automated-decision transparency makes provenance packets material.

Policy: Breadth claims ('stakeholders uniformly support', 'jurisdictions report materially higher...') are common and dangerous. Volatility often high because the policy area is contested by design.

6. Integration With Elemental Agents / LegiPro Runtime

Source-routed baseline → produces candidate. Accountant red-team lens → challenges exactly the five dimensions. Deterministic selector → applies professional-liability weights (Scorecard is the visible version of that logic). Provenance packet → Usage Log + repair worksheets make later replay or external review possible.

7. Limitations

The numeric index is an ordinal risk signal, not a probability. A LOW score does not guarantee truth; a CRITICAL score does not prove falsehood. Weights are pilot-derived on accounting/tax/administrative dossiers. Different domains may justify re-weighting after local validation. Assumes a human reviewer with domain literacy. Scores individual claims/passages.

8. Versioning & Deliverables

v1.0 — June 2026 (tied to LegiPro Mirofish public provenance). Lifetime updates for purchasers. Files: AI_Hallucination_Risk_Scorecard_v1.xlsx (6 sheets), this Methodology PDF, quick-reference card, sector one-pagers.

PHAROS AI — Elemental Agents. Public provenance: github.com/martinlepage26-bit/pharos-suite (docs/evidence/2026-06-05-legipro-mirofish/).